School of Engineering and Applied Science (SEAS), Ahmedabad University

CSE 400: Fundamentals of Probability in Computing

Project Scribe Report (Questions 1-3)

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Scribe Question 1: Project System and Objective
What is the probabilistic problem being addressed in your project? Clearly state the system objective and identify the primary sources of uncertainty involved.

Answer:

1. Probabilistic Problem Being Addressed

The main probabilistic problem in our Device-to-Device (D2D) Edge Computing project is the **uncertainty in task processing time**, meaning that the exact time required to execute a task is not known beforehand.

Most existing approaches assume that this time is fixed or deterministic. However, in real systems, task execution time varies depending on factors such as input data size and task complexity. When the system relies on an average execution time, the arrival of a heavy task can occupy resources longer than expected, leading to congestion and missed deadlines for other tasks.

2. Understanding the Problem (Highway Analogy)

> To visualize this, consider a **highway toll booth**: if the booth assumes every vehicle takes exactly 10 seconds to pass, the arrival of a large truck that takes 60 seconds will create a traffic jam for all cars behind it. Similarly, in our network, if a heavy task is treated like a normal one, it blocks the processing queue and delays other tasks. This analogy highlights why a probabilistic approach is necessary: the system must account for variability in task execution rather than relying on fixed estimates.

3. System Objective

The objective of our system is to **maximize the Task Completion Rate**, which is the proportion of tasks completed within their deadlines. To achieve this, we propose a **game-based probabilistic offloading scheme**.

Instead of assuming exact execution times, each device estimates the **probability of successful task completion** before offloading. Game theory, particularly **Nash Equilibrium**, ensures that devices make balanced decisions and prevents overloading any single node.

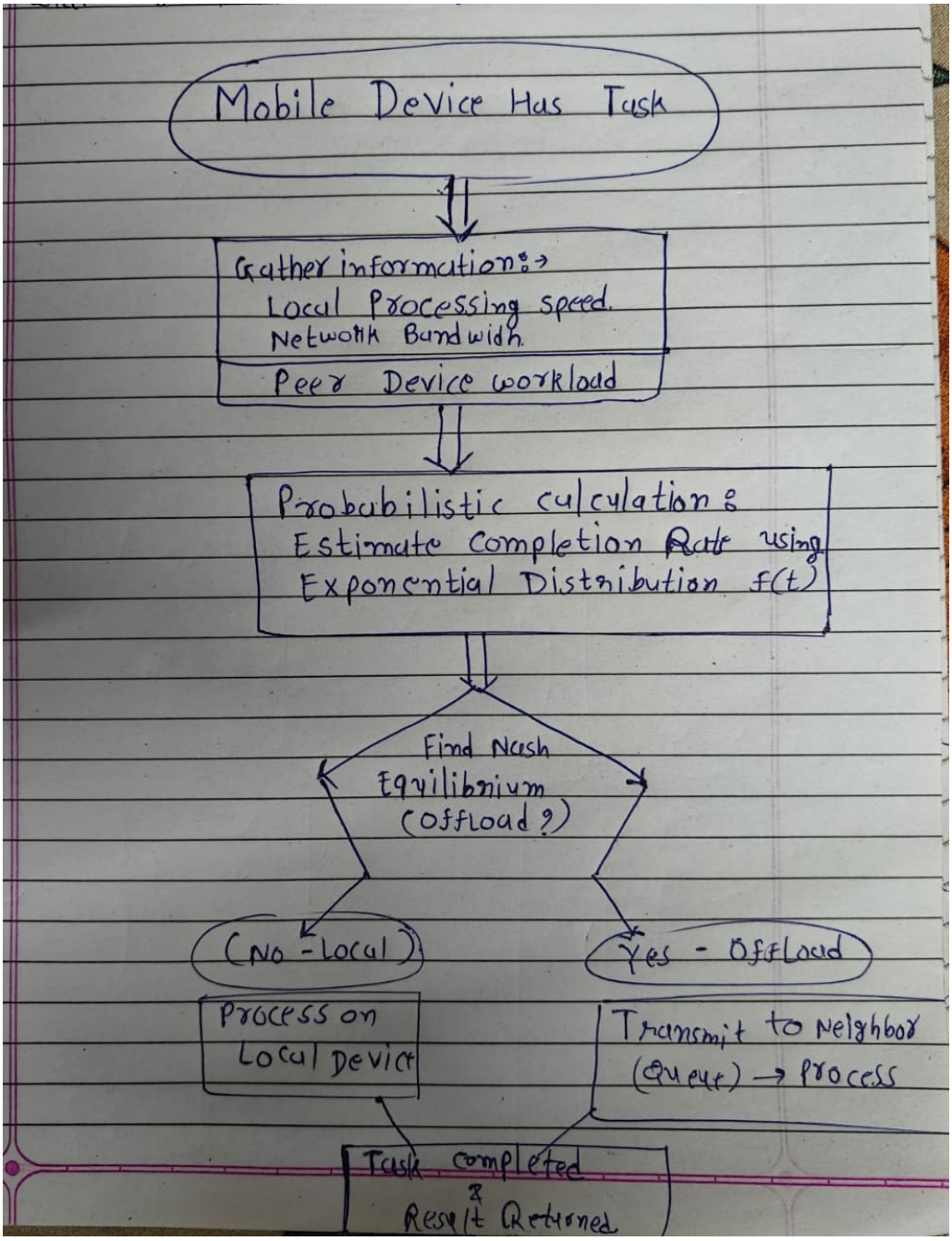

Figure 1: System Flow Process: Decision Making for Task Offloading

4. Primary Sources of Uncertainty

- **Task Processing Time (TPT):** Varies due to data-dependent execution, different application inputs, and varying execution paths. For example, processing an image with more features takes longer than a simple image.
- **Queueing and Contention Delay:** Multiple tasks competing for limited resources can lead to unpredictable waiting times.
- **Device Heterogeneity:** Differences in computational capabilities and workloads across devices cause variability in task execution.
- **Transmission Delay:** Offloading introduces communication delays that fluctuate depending on network conditions.

To handle these uncertainties, particularly TPT, our system uses a **probabilistic model (exponential distribution)**, which allows more realistic scheduling decisions.

Scribe Question 2: Key Random Variables and Uncertainty Modeling
Identify the key random variables in your project and describe how uncertainty is modeled for each. Clearly state any probabilistic assumptions currently being made.

Answer:

1. Key Random Variables

In our project, several system components are modeled as random variables because their values cannot be predicted exactly in advance.

- **Task Processing Time (t):** Modeled as a continuous random variable. Its value depends on the application type and input data size. Since different applications show different execution behaviors, application-specific probability distributions are used.
- **Task Arrival Process:** Modeled as a Poisson random process with arrival rate λ_k. This reflects that tasks are generated by users at random times, and exact arrival instants are unknown.
- **Task Completion Time:** A random variable that includes queue waiting time, processing time, and transmission delay. Since each component is uncertain, the overall completion time is random.
- **Task Completion Indicator:** Modeled as a Bernoulli random variable indicating whether a task is completed before its deadline (1 if met, 0 otherwise). This is directly related to system performance.
- **Device Workload (λ_j):** Represents the arrival rate of tasks at a particular device. Since task arrivals fluctuate, workload varies dynamically.

2. Uncertainty Modeling

Uncertainty in Task Processing Time is modeled using measurement-based probabilistic distributions, such as **Exponential**, Gaussian, or Gamma distributions.

- The choice depends on the observed behavior of applications.
- **Offline profiling** determines the distribution family based on historical data.
- **Online parameter calibration** updates parameters using runtime samples, making the model scalable.

3. Current Probabilistic Assumptions

- **Independent Arrivals:** Task arrivals are independent; the arrival of one task does not affect another.
- **i.i.d. Service Times:** Service times for tasks of the same type are independent and identically distributed.
- **Stable Short-term Dynamics:** System dynamics are assumed relatively stable during decision-making for reliable estimation.

Scribe Question 3: Probabilistic Reasoning and Dependencies
Explain how probabilistic relationships (e.g., dependence, independence, conditionality) are used within your project system to support reasoning, inference, or decision-making.

Answer:

In our project, probabilistic reasoning is central to how devices make offloading decisions. The system accounts for dependence, conditionality, and independence to model interactions.

1. Dependence Between Devices

Devices in a D2D network are interdependent because they share computational resources. When one device offloads a task to another, it increases the workload and queue length of the target device. This directly affects the probability that subsequent tasks will meet their deadlines.

- Our system uses **Game-Theoretic reasoning**, where each device considers congestion effects caused by others.
- Performance emerges from collective behavior rather than independent actions.

2. Conditional Probability for Decision-Making

The offloading algorithm uses conditional probability to make informed decisions. Instead of checking if a device is simply "fast," the system evaluates:

$$P(\text{Success} \mid \text{Target} = j)$$

This probability depends on the current queue length, processing rate, and workload of the target device j. By conditioning on the target, the system avoids congested nodes.

- *Example:* Processing a photo with 1 face is fast, while 20 faces take longer. The system uses probabilistic modeling (e.g., exponential distribution) to estimate success probabilities based on these conditions.

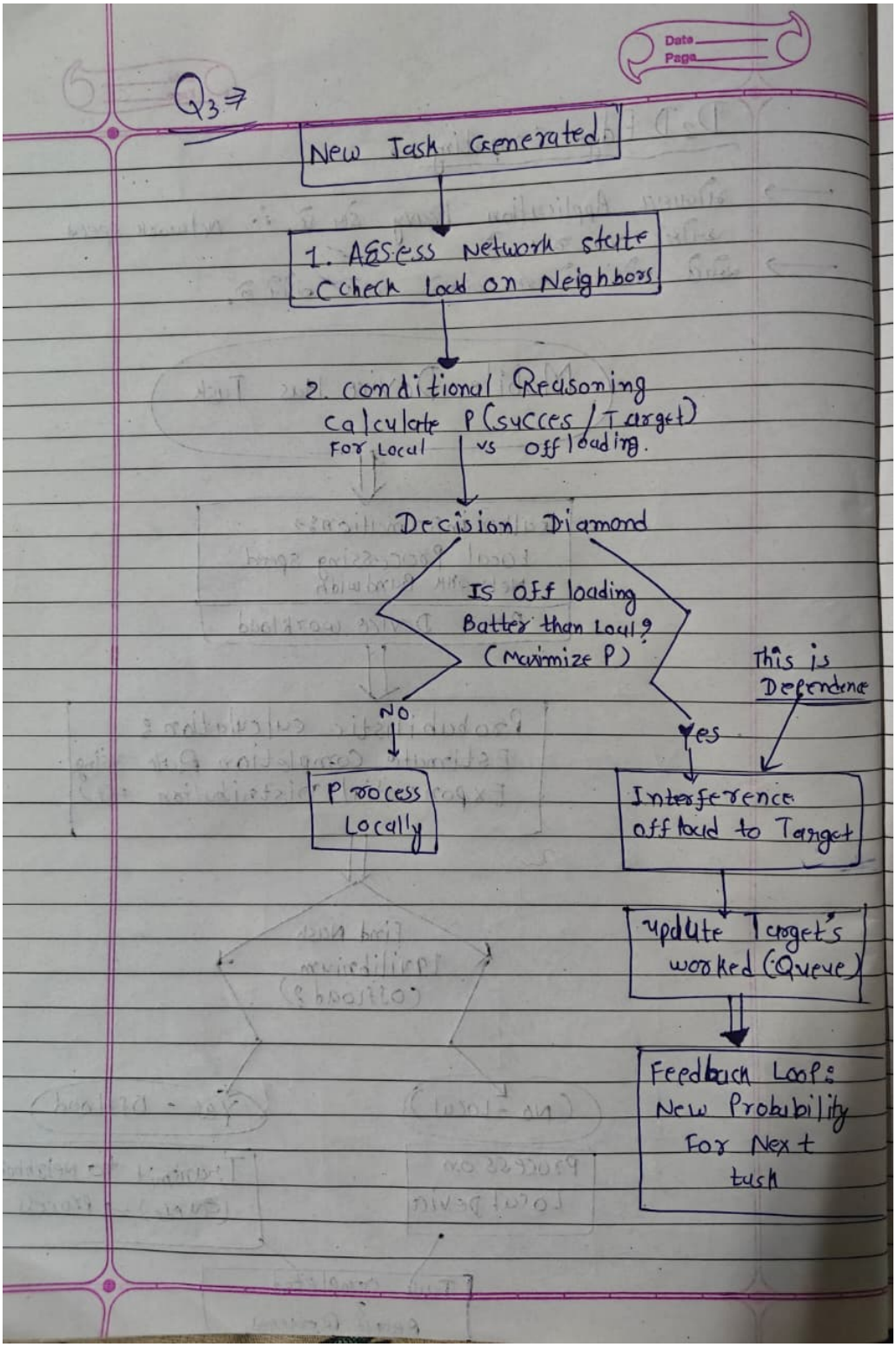

Figure 2: Probabilistic Reasoning Loop: Dependence and Decision Logic

3. Independence Assumptions to Simplify Analysis

To make the system computationally efficient for real-time use:

- **Task arrivals are independent:** Modeled as a Poisson process.

- **Memoryless Property:** Due to the Exponential distribution assumption, the remaining processing time does not depend on how long a task has already been running.

These assumptions allow the use of the **M/G/1 queueing model** to derive completion probabilities from the CDF.

4. Decision-Making and Nash Equilibrium

Since devices affect one another, the system iteratively updates offloading decisions until a **Nash Equilibrium** is reached. At this point, each device has selected the best possible strategy given the choices of others, and no device can improve its outcome by changing its decision alone.

End of Scribe Report Submission